CEHD
COLLEGE OF EDUCATION
+ HUMAN DEVELOPMENT
UNIVERSITY OF MINNESOTA

M. Ed. / Initial Licensure in Science Education: Grades 9-12 Life Science
Prerequisite Course Evaluation

Candidates are required to have an undergraduate degree from an accredited institution with a major in **life science <u>or</u>** closely related field, or meet the prerequisites below, and with appropriate liberal arts/general education coursework. Courses in each of the content areas are required to meet Standard 1 of MNSEPT: "*The learner understands and can create learning experiences that make these aspects of content matter meaningful for students" (https://www.revisor.mn.gov/rules/?id=8710.2000).* The X's in the "Course/Number" column reflect outstanding prerequisites. Any in-progress coursework would be counted as complete if an official transcript with proof of enrollment is provided at the time of application.

Candidate Name:	**UMN ID#:**	**Email:**	**Date:**
Degree Granting Institution:	**Major(s):**	**Minor(s):**	**Prepared by:**

Life Science Major Status

Requirements	***Major***	**Degree Granting Institution/ Degree Year**	**Life Science Degree Requirement Check ✔**
• An undergraduate life science (biology) degree from an accredited institution in the U.S. or an equivalent 4 year life science undergraduate degree from an accredited international institution.			

University of Minnesota, Twin Cities students must meet the following course requirements. If non-U of M candidates do not hold a life science degree, they also must meet the course requirements below.

Content Area	**Requirement**	**U of M Course Options**	**Course/Term/Year**	**Credits /Grade**	**Institution If Other Than Degree Granting**
Psychology	General Psychology (4 credits)	PSY 1001			
Life Science	Biology I w/ Lab (4 credits)	BIOL 1001, 1009, or 1951/1961			
	General Zoology (4 credits)	BIOL 2012, 2005/3211, or EEB 3412W			

Additional Life Science courses (eight courses)	General Botany (3 credits)	BIOL 2022, PMB 3007W, or 3002/3005W			
	Physiology (3 credits)	PHSL 3051 or BIOL 3211			
	Ecology (3 credits)	EEB 3807, 3407, or 3408W			
	Evolution (3 credits)	EEB 3409			
	Biochemistry (3 credits)	BIOC 3021 or 3022			
	Genetics (3 credits)	GCD 3022 or BIOL 4033			
	Cell biology (3 credits)	GCD 3033, 4005W, or BIOL 4004			
Research in the Life Sciences	(2 credits)	BIOL 4994/4794W, BIOC 4994/4794W, COP 4994/4794W, EEB 4994/4794W, FW 4108, GCD 4994/4794W, MICB 4994/4794W, NSCI 4994/4794W, PMB 4994/4794W			
Life Science Electives, if needed					

CANDIDATES WITHOUT A SCIENCE MAJOR MUST MEET COURES FOR AN EQUIVALENT OF AN UNDERGRADUATE MAJOR. Per U of MN program requirements, 39 credits are required for a Life Science degree (including General Psychology). Adjustment can be made for Non U of M degree programs with varying credit requirements.

Additional Licensure in Grades 5-8 General Science (Optional)
Prerequisite Course Evaluation

Content Area	Requirement	U of M Course Options	Course/Term/Year	Credits/Grade	Institution
Life Science two courses (with labs)	Biology I	BIOL 1001 or 1009 or 1951/1961			
	Biology II	BIOL 2003/3004 or 1055 or GCD 3022 or 3033			
Physics sequence two courses (with labs)	Physics I	PHYS 1101 or 1221 or 1301W			
	Physics II	PHYS 1102 or 1222 or 1302W			
Chemistry sequence two courses (with labs)	General Chemistry I	CHEM 1061 and 1065 or 1081 and 1065			
	General Chemistry II	CHEM 1062 and 1066 or 1082 and 1086			
Earth science two courses (with labs)	Physical Geology	ESCI 1001			

	Astronomy	AST 1001			